Descriptive Statistics from Generated Data

Lecture Notes

General notes for introductory probability and data analysis

These notes introduce the general mathematical and statistical ideas needed to move from probability models to observed data. The aim is to understand how data are generated, summarized, visualized, compared, and interpreted.

Contents

1 From Probability Models to Data

In earlier parts of probability theory, we usually begin with a mathematical model. We define a sample space, a probability measure, and perhaps a random variable. We then compute probabilities from the model.

In descriptive statistics the direction is different. We begin with observations. These observations may have been collected from the real world, or they may have been generated by a simulation. The central question is not only

What does the model say?

but also

What do the observed data show?

Descriptive statistics is the part of statistics concerned with organizing, summarizing, visualizing, and interpreting data. It does not require a full formal probability model, but it often uses probabilistic language such as empirical probability, relative frequency, random sample, and distribution.

Descriptive statistics

Descriptive statistics consists of methods used to describe a dataset by means of tables, numerical summaries, and visualizations. Its goal is to make the structure of the data understandable.

The word *descriptive* is important. A descriptive statistic does not automatically prove a causal relation, establish a universal law, or guarantee that the same pattern will occur in another sample. It tells us what is visible in the data we have.

2 Datasets, Observations, and Variables

A dataset is usually arranged as a rectangular table. Each row represents an observation, and each column represents a variable.

Observation

An observation is one recorded unit in a dataset. Depending on the context, an observation may be a person, a delivery, a product, a transaction, a measurement, an hour, a customer, or a simulated trial.

Variable

A variable is a recorded characteristic of an observation. Examples include age group, score, delivery time, machine identifier, number of calls, satisfaction level, or whether a customer renewed a subscription.

2.1 Example of a data table

observation_id	group	time_min	score	status
A001	morning	28.4	82	completed
A002	evening	41.7	76	completed
A003	morning	35.2	91	completed
A004	evening	64.9	58	delayed

Here one row is one observation. The variables are `group`, `time_min`, `score`, and `status`. Some variables are numerical, while others are categorical.

2.2 Types of variables

Categorical variable

A categorical variable records membership in a category. Examples: device type, customer channel, machine, zone, payment method.

Numerical variable

A numerical variable records a number on which arithmetic operations have meaning. Examples: delivery time, exam score, length, number of requests, waiting time.

Categorical variables may be *nominal* or *ordinal*.

- A nominal variable has categories without a natural order, such as `website`, `phone`, `mobile app`.
- An ordinal variable has ordered categories, such as satisfaction levels from 1 to 5 or ratings `low`, `medium`, `high`.

Numerical variables may be *discrete* or *continuous*.

- A discrete variable takes separated values, often counts: number of requests, number of defects.
- A continuous variable is measured on a scale: time, length, mass, temperature.

A common mistake

A number in a dataset is not automatically a numerical variable in the statistical sense. Postal codes, student identifiers, and product IDs may be written as numbers, but averaging them is meaningless.

3 Samples, Populations, and Reproducible Generation

3.1 Population and sample

Population

The population is the full collection of units about which we would like to say something.

Sample

A sample is the collection of units actually observed. In practice, we often have access only to a sample, not the entire population.

For example, if a company wants to understand delivery times for all deliveries made in a month, then all those deliveries form the population. If we analyze only 500 selected deliveries, those 500 deliveries form a sample.

3.2 Random samples and sample variability

A random sample is not a perfect copy of the population. Different random samples from the same process will usually give different means, different histograms, different frequencies, and different outliers.

Important idea

Randomness does not disappear because we use a fixed sample. A fixed dataset is only one realization of a random process. Another realization may look similar in general structure but different in details.

3.3 Random seeds and reproducibility

In simulations, a random seed initializes the pseudo-random number generator. If the same algorithm, same software, and same seed are used, the same sequence of generated values can often be reproduced.

Random seed

A random seed is a value used to initialize a pseudo-random number generator. It is a tool for reproducibility, not a mathematical proof that the generated dataset is unique or inevitable.

The same data-generating mechanism can produce many different samples when the seed changes. Comparing results across several seeds helps distinguish stable patterns from random fluctuations.

4 Frequency Tables and Empirical Distributions

4.1 Absolute frequencies

Suppose a categorical variable X can take values $a_1, a_2, \dots, a_k$. If the dataset contains n observations, the absolute frequency of category a_j is

$$n_j = \#\{i : x_i = a_j\}.$$

The numbers $n_1, \dots, n_k$ form a frequency table. They satisfy

$$n_1 + n_2 + \cdots + n_k = n.$$

4.2 Relative frequencies

The relative frequency of category a_j is

$$f_j = \frac{n_j}{n}.$$

Relative frequencies satisfy

$$f_1 + f_2 + \cdots + f_k = 1.$$

They can be interpreted as empirical probabilities: probabilities estimated from observed data.

Empirical probability

For an event A observed in a dataset of size n, the empirical probability of A is

$$\widehat{P}(A) = \frac{\#\{\text{observations satisfying } A\}}{n}.$$

4.3 Example: frequency table

Example with an independent context

A small dataset records the preferred payment method of 50 customers.

payment method	absolute frequency	relative frequency
card	24	0.48
cash	11	0.22
transfer	9	0.18
mobile	6	0.12

The empirical probability that a randomly selected observed customer used a card is $24/50 = 0.48$.

4.4 Bar charts

Bar charts are used mainly for categorical or discrete variables with a small number of possible values.

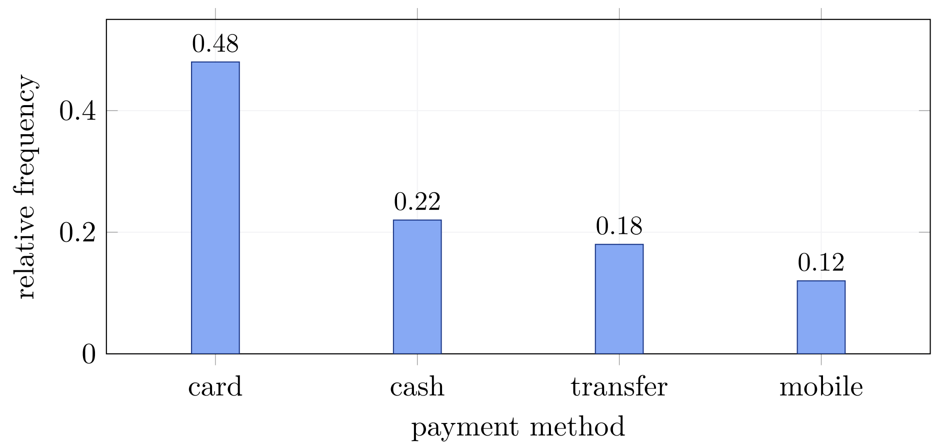


Figure 1: A bar chart represents relative frequencies of categories.

5 The Empirical Distribution Function

For numerical data $x_1, x_2, \ldots, x_n$, the empirical cumulative distribution function, or empirical CDF, is defined by

$$\widehat{F}_n(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{\{x_i \le x\}}.$$

It gives the proportion of observations less than or equal to x.

Empirical CDF

The empirical CDF at x is the fraction of observations that are not greater than x.

$$\widehat{F}_n(x) = \frac{\#\{i : x_i \le x\}}{n}.$$

The empirical CDF is a step function. It increases only at observed values. If many observations are less than or equal to a threshold, the empirical CDF at that threshold is large.

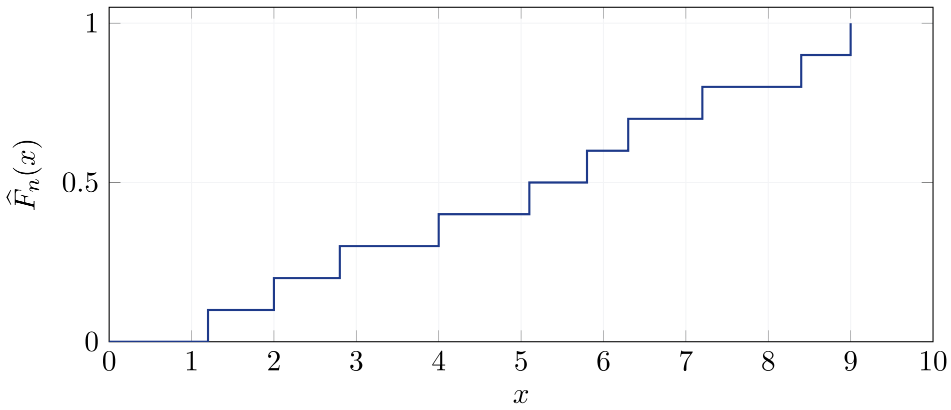


Figure 2: An empirical CDF is a step function.

6 Measures of Location

Measures of location describe where the data are centered.

6.1 Mean

For numerical observations $x_1, \ldots, x_n$, the sample mean is

$$\overline{x} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

The mean uses every observation. It is natural when values combine additively, such as total time, total cost, total length, or total score.

Example

If five processing times are

$$12, \quad 15, \quad 13, \quad 20, \quad 40,$$

then

$$\overline{x} = \frac{12 + 15 + 13 + 20 + 40}{5} = 20.$$

The value 40 strongly affects the mean.

6.2 Median

The median is the middle value after sorting the data. If n is odd, it is the central observation. If n is even, it is usually defined as the average of the two central observations.

Median

The median is a value that divides ordered data into two parts: at least half of the observations are less than or equal to it, and at least half are greater than or equal to it.

The median is often more robust than the mean when the data are skewed or contain outliers.

Mean versus median

When a distribution is strongly right-skewed, the mean is often larger than the median. This does not mean either number is wrong. They answer different questions.

6.3 Mode

The mode is the most frequent value or category. It is especially useful for categorical variables.

7 Measures of Variability

Two datasets can have the same center but very different spread. Measures of variability describe how much observations differ from each other.

7.1 Range

The range is

$$\max(x_i) - \min(x_i).$$

It is easy to compute but depends only on two observations.

7.2 Sample variance and standard deviation

The sample variance is

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2.$$

The sample standard deviation is

$$s = \sqrt{s^2}.$$

Standard deviation

The standard deviation measures a typical size of deviation from the mean. It has the same unit as the original variable.

The denominator $n-1$ is used when the data are treated as a sample from a larger population. If the dataset is the entire population under study, the population variance uses denominator n.

Interpretation

A larger standard deviation means the observations are more dispersed around the mean. It does not by itself say whether values are good or bad; interpretation depends on context.

7.3 Why squared deviations?

The deviations $x_i - \overline{x}$ sum to zero:

$$\sum_{i=1}^{n}(x_i - \overline{x}) = 0.$$

To measure spread, we need to avoid cancellation. Squaring deviations makes them nonnegative and gives more weight to large deviations.

8 Quartiles, IQR, and Outliers

8.1 Quartiles

Quartiles divide ordered data into four parts.

- Q_1 is the first quartile, near the 25th percentile.
- Q_2 is the second quartile, equal to the median.
- Q_3 is the third quartile, near the 75th percentile.

8.2 Interquartile range

The interquartile range is

$$\mathrm{IQR} = Q_3 - Q_1.$$

It measures the spread of the middle half of the data.

Interquartile range

The interquartile range is the distance between the third and first quartiles. It is less sensitive to extreme observations than the full range or standard deviation.

8.3 The 1.5 IQR rule

A common descriptive rule labels observations as possible outliers if they fall outside

$$[Q_1 - 1.5\,\mathrm{IQR},\ Q_3 + 1.5\,\mathrm{IQR}].$$

Outliers require interpretation

An outlier is not automatically an error. It may be a measurement problem, a rare but valid event, a special subgroup, or a signal of a different process.

9 Histograms and Distribution Shape

A histogram is used for numerical data. It divides the numerical axis into intervals called bins and counts how many observations fall into each bin.

Histogram

A histogram represents the distribution of a numerical variable by grouping values into intervals and showing frequencies or relative frequencies for those intervals.

Histograms help identify:

- approximate center,
- spread,
- skewness,
- multiple peaks,
- gaps,
- possible outliers.

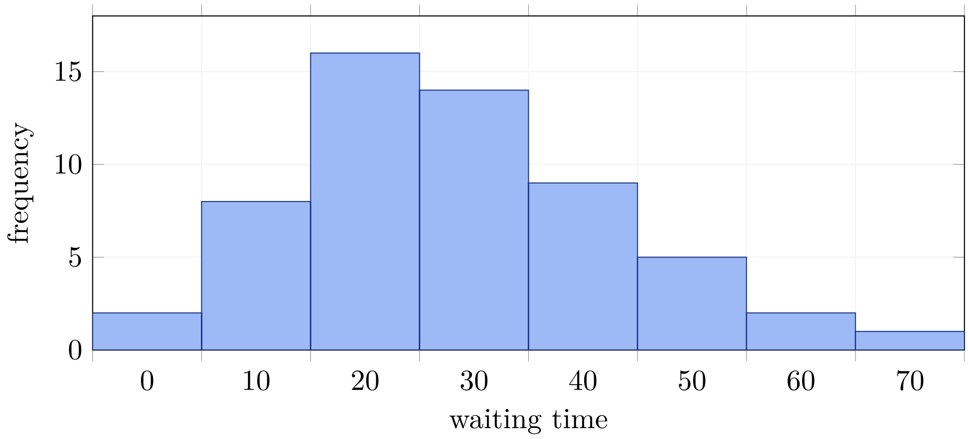


Figure 3: A histogram for numerical data. The choice of bins affects the visual impression.

9.1 Skewness

A distribution is roughly symmetric if the left and right sides are similar. It is right-skewed if it has a long right tail, and left-skewed if it has a long left tail.

Mean and skewness

In a right-skewed distribution, a few very large values pull the mean to the right. The median often better represents a typical observation.

10 Boxplots

A boxplot summarizes a numerical distribution using quartiles and possible outliers.

A standard boxplot shows:

- the median,
- the first and third quartiles,
- the interquartile range,
- whiskers extending to non-outlying values,
- possible outliers as separate points.

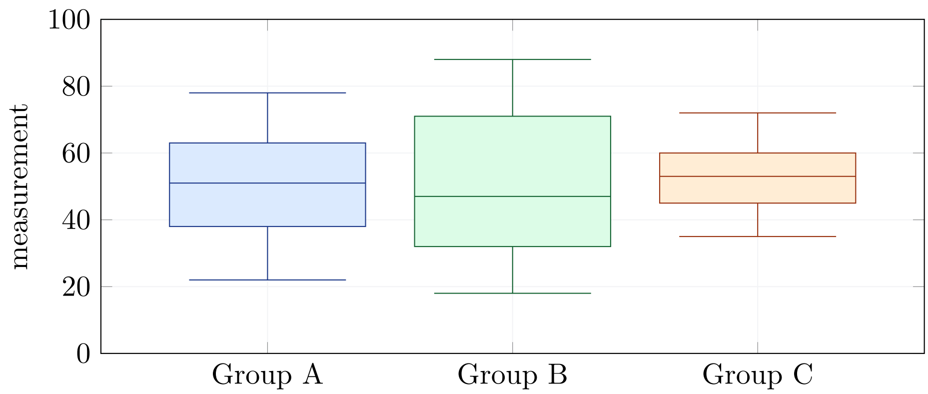


Figure 4: Boxplots make group comparisons easier.

Boxplots are compact, but they hide details of the distribution. They should often be combined with histograms, dot plots, or empirical CDFs.

11 Comparing Groups

Many datasets contain a grouping variable. Examples include machines, experimental groups, customer channels, delivery zones, departments, or time periods.

When comparing groups, it is usually not enough to compare only means. A responsible comparison should consider:

- sample size in each group,
- center: mean and median,
- variability: standard deviation, IQR, range,
- shape: symmetry, skewness, multimodality,
- outliers,
- context-specific rates or proportions.

Example: two groups with the same mean

Suppose two machines both produce parts with mean length close to 50 mm. Machine A has standard deviation 0.2 mm, while Machine B has standard deviation 1.1 mm. Even with the same mean, Machine B may produce many more parts outside specification limits.

Comparing only averages

A higher mean does not always mean better performance. In scores, higher may be better. In delivery time, lower may be better. In manufacturing, being close to a target may be more important than being high or low.

12 Empirical Probabilities and Conditional Frequencies

Descriptive statistics often connects naturally with probability. If a dataset records whether each customer renewed a subscription, the observed renewal rate is an empirical probability.

12.1 Empirical probability of an event

If A is a property of an observation, then

$$\widehat{P}(A) = \frac{n(A)}{n}.$$

For example, if 180 out of 600 customers renewed, then

$$\widehat{P}(\text{renewed}) = \frac{180}{600} = 0.30.$$

12.2 Conditional empirical probability

If B is another property and $n(B) > 0$, then

$$\widehat{P}(A \mid B) = \frac{n(A \cap B)}{n(B)}.$$

Example

Among 120 customers using a mobile app, 54 renewed. The empirical conditional probability of renewal among mobile app customers is

$$\widehat{P}(\text{renewed} \mid \text{mobile app}) = \frac{54}{120} = 0.45.$$

Conditional probabilities are directional

In general,

$$\widehat{P}(A \mid B) \neq \widehat{P}(B \mid A).$$

The first asks for the proportion of B-observations that also satisfy A. The second asks for the proportion of A-observations that also satisfy B.

13 Scatter Plots, Covariance, and Correlation

When a dataset contains two numerical variables measured on the same observations, scatter plots and correlation are basic tools.

13.1 Scatter plots

A scatter plot displays pairs (x_i, y_i). It allows us to see direction, strength, curvature, clusters, and outliers.

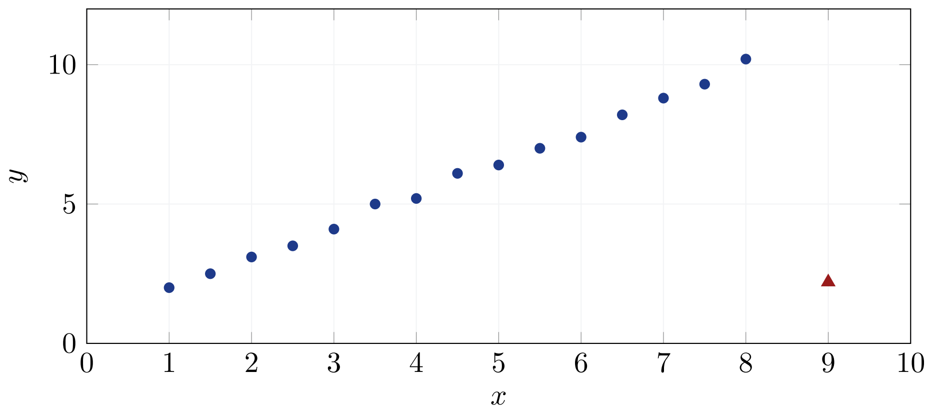


Figure 5: A scatter plot shows relationships and possible outliers.

13.2 Covariance

The sample covariance of two numerical variables is

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \overline{x})(y_i - \overline{y}).$$

If large values of x tend to occur with large values of y, the covariance is positive. If large values of x tend to occur with small values of y, it is negative.

Covariance depends on units. If we change centimeters to meters, the covariance changes numerically.

13.3 Correlation coefficient

The sample correlation coefficient is

$$r = \frac{s_{xy}}{s_x s_y}.$$

It always satisfies

$$-1 \leq r \leq 1.$$

Correlation coefficient

The correlation coefficient measures the strength and direction of a linear relationship between two numerical variables.

- r close to 1 means a strong positive linear relationship.
- r close to -1 means a strong negative linear relationship.
- r close to 0 means no strong linear relationship.

13.4 Correlation traps

Correlation does not see everything

A low correlation does not necessarily mean no relationship. The relationship may be nonlinear. Also, a high correlation does not prove causation.

Outliers can strongly distort correlation. A single unusual point may increase, decrease, or even reverse the apparent linear association.

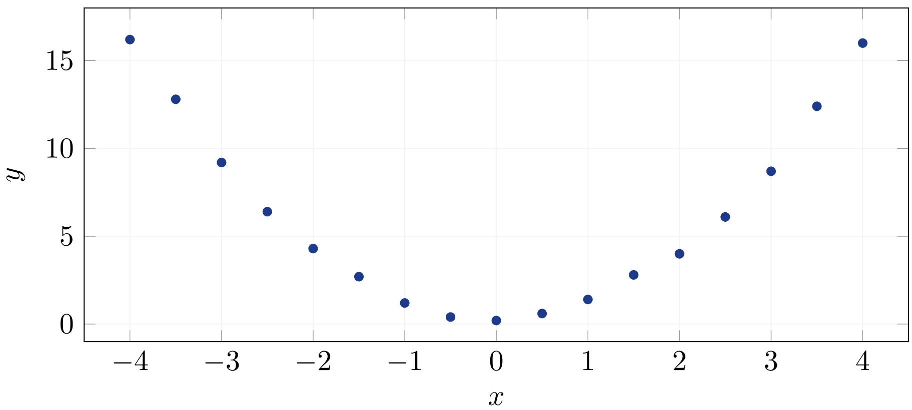


Figure 6: A nonlinear relationship may have low linear correlation.

14 Time, Order, and Cumulative Patterns

Some data are naturally ordered: trials, days, hours, weeks, or transactions over time. In such cases, cumulative summaries are useful.

For a sequence of indicator variables $I_1, I_2, \ldots, I_n$, where $I_i = 1$ if an event occurs and $I_i = 0$ otherwise, the cumulative count after t observations is

$$C_t = \sum_{i=1}^{t} I_i.$$

The cumulative relative frequency is

$$R_t = \frac{C_t}{t}.$$

Stabilization of relative frequencies

For small t, the cumulative relative frequency may fluctuate strongly. As t grows, it often becomes more stable, especially when observations are generated from the same probability mechanism.

This idea is closely related to the law of large numbers, but in descriptive statistics we can observe it directly by plotting cumulative relative frequencies.

15 Generated Data and Simulation

Generated datasets are useful for learning because the data-generating mechanism is known. We can compare what we observe with what we know was used to generate the data.

A typical workflow is:

1. define a data-generating process,
2. set a seed for reproducibility,
3. generate data,
4. save the dataset,
5. analyze it as observed data,
6. repeat with different seeds,
7. compare which conclusions are stable.

Listing 1: Generic simulation pattern

```
import numpy as np
import pandas as pd

rng = np.random.default_rng(123)
n = 1000

x = rng.normal(loc=10, scale=2, size=n)
df = pd.DataFrame({"observation_id": range(1, n + 1), "x": x})

df.to_csv("generated_data.csv", index=False)
```

Reproducible does not mean universal

A seed makes a simulated dataset reproducible. It does not mean the observed sample exactly equals the theoretical distribution, and it does not mean another seed would give identical summaries.

16 Practical Data Analysis Workflow

A good descriptive analysis should follow a logical sequence.

16.1 Step 1: Understand the rows and variables

Ask:

- What does one row represent?
- What is the unit of observation?

- Which variables are categorical?
- Which variables are numerical?
- Are there grouping variables?

16.2 Step 2: Check basic structure

Useful checks include:

```
df.head()
df.info()
df.describe()
df["group"].value_counts()
```

These commands do not replace interpretation. They only reveal the structure.

16.3 Step 3: Summarize categorical variables

Use frequency tables and bar charts.

16.4 Step 4: Summarize numerical variables

Use mean, median, standard deviation, quartiles, histograms, boxplots, and empirical CDFs.

16.5 Step 5: Compare groups

Group comparisons may use summaries such as:

```
df.groupby("group")["value"].describe()
```

But the result must be interpreted. A table alone is not a conclusion.

16.6 Step 6: Study relationships

For two categorical variables, use cross-tabulations and conditional frequencies. For two numerical variables, use scatter plots, covariance, and correlation.

17 What Can Go Wrong?

Descriptive statistics is powerful, but it is easy to misuse.

17.1 Mistaking summaries for explanations

A numerical summary describes the data. It does not automatically explain why the data look that way.

17.2 Ignoring sample size

A proportion based on 5 observations is much less stable than a proportion based on 500 observations.

17.3 Using the wrong plot

A bar chart is appropriate for categorical data. A histogram is appropriate for numerical data. A line plot should usually be used when order or time matters.

17.4 Ignoring outliers

Outliers may dominate means, standard deviations, and correlations.

17.5 Confusing correlation with causation

Correlation alone does not establish a causal mechanism. Additional design, assumptions, or experimental evidence are needed.

17.6 Reporting numbers without context

A statistic should be connected to the meaning of the variable. For example, a standard deviation of 3 minutes may be small in long-distance shipping but large in a fast-response emergency system.

18 Writing a Short Statistical Report

A statistical report should combine numbers, plots, and interpretation.

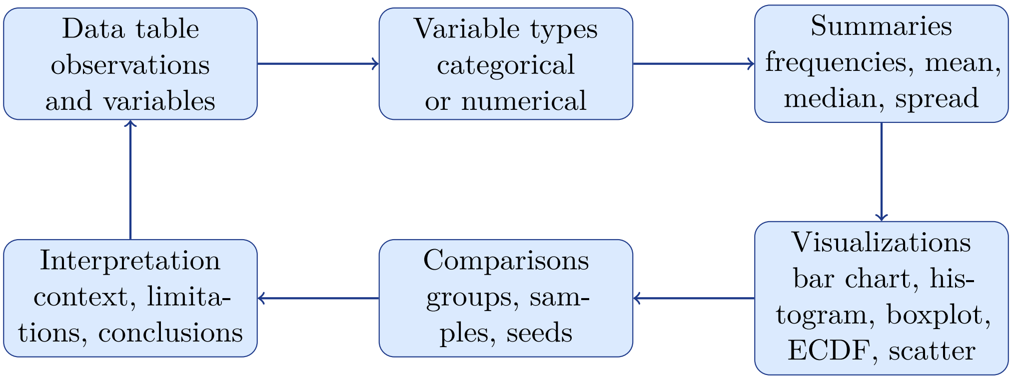
A clear report usually contains:

1. description of the data and unit of observation,
2. number of observations and variables,
3. classification of variables,
4. frequency tables for categorical variables,
5. numerical summaries for numerical variables,
6. appropriate plots,
7. comparisons between groups if groups exist,
8. empirical probabilities or conditional frequencies when relevant,
9. discussion of uncertainty or sample-to-sample variation,
10. warnings about possible misleading summaries,
11. a short conclusion in words.

A good conclusion

A good conclusion does not simply repeat numbers. It explains what the numbers and plots suggest in the context of the data, and it mentions important limitations.

19 Concept Map



20 Summary

Descriptive statistics begins with data. A dataset must first be understood structurally: what the rows mean, what the variables mean, and what types of variables are present. Categorical variables are summarized by frequencies, relative frequencies, and bar charts. Numerical variables are summarized by means, medians, variances, standard deviations, quartiles, IQRs, histograms, boxplots, and empirical CDFs.

A single statistic rarely tells the whole story. Means can be distorted by outliers. Correlations may miss nonlinear relationships. Standard deviations may hide group structure. Plots often reveal patterns that tables alone do not show.

Generated data are especially useful for learning because one can compare repeated samples from the same mechanism. Changing the random seed usually changes the sample. A good analysis therefore separates stable patterns from random sample-to-sample fluctuations.

The final goal is not mechanical calculation. The final goal is interpretation: to connect summaries and visualizations with the real or simulated situation represented by the data.